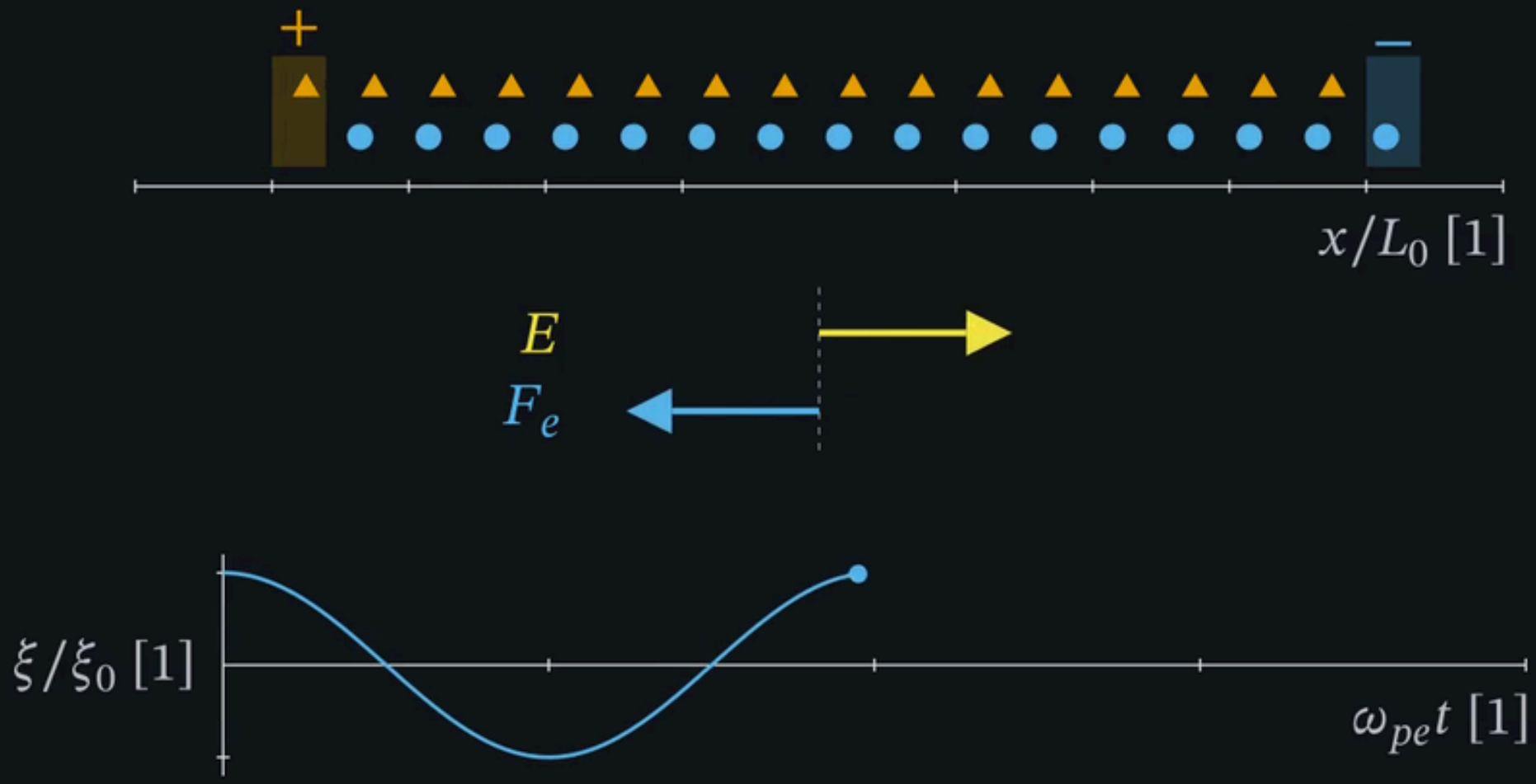
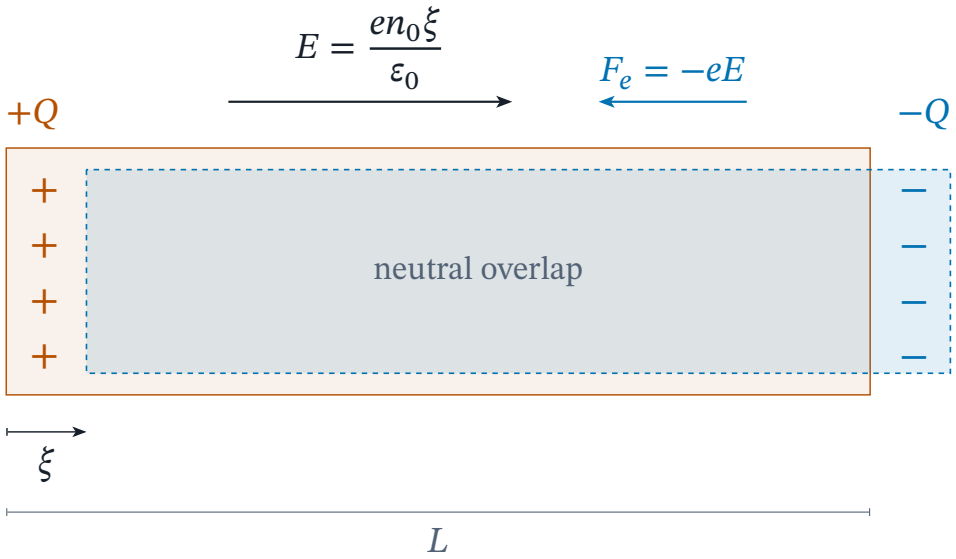
Plasma oscillations

Christopher Albert


+
−
x/L_0 [1]
E
F_e
ξ/ξ_0 [1]
$\omega_{pe}t$ [1]

3.1 Electron plasma oscillations



$Q = en_0A\xi$ A slab area

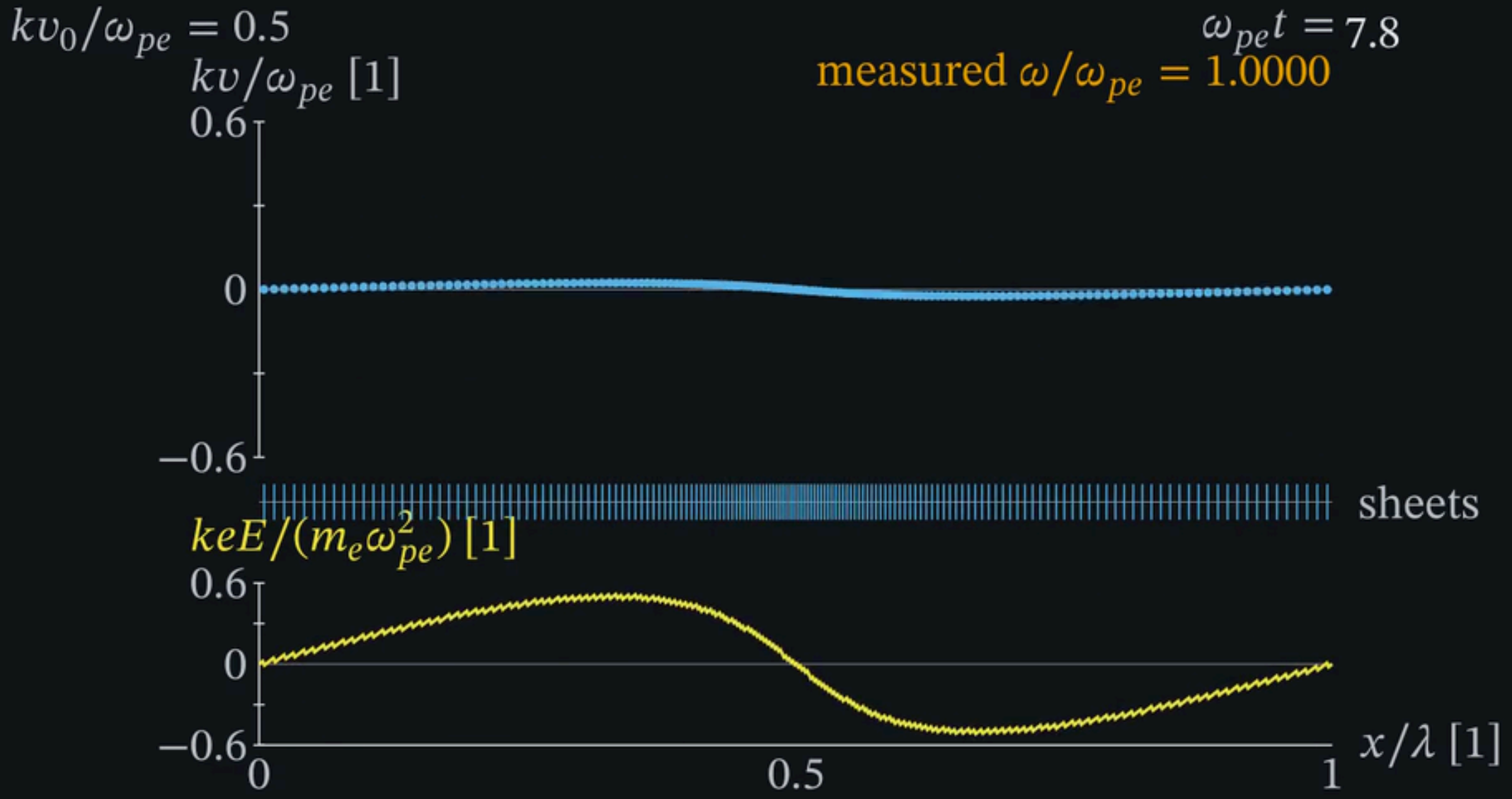
Gauss $\varepsilon_0 EA = Q$

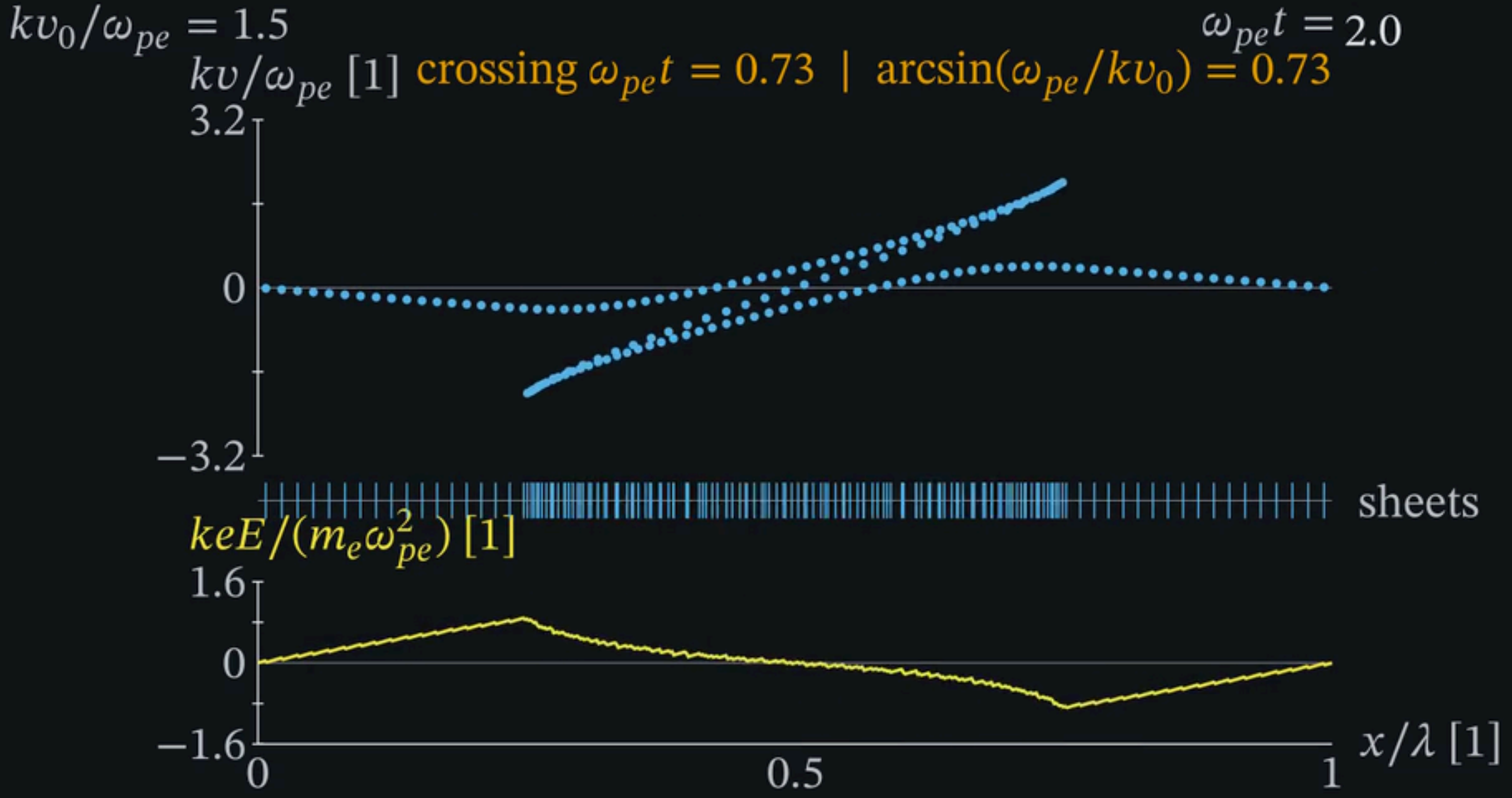
Newton $m_e \frac{\mathrm{d}^2\xi}{\mathrm{d}t^2} = -eE$

$$\frac{\mathrm{d}^2\xi}{\mathrm{d}t^2} + \omega_{pe}^2\xi = 0$$

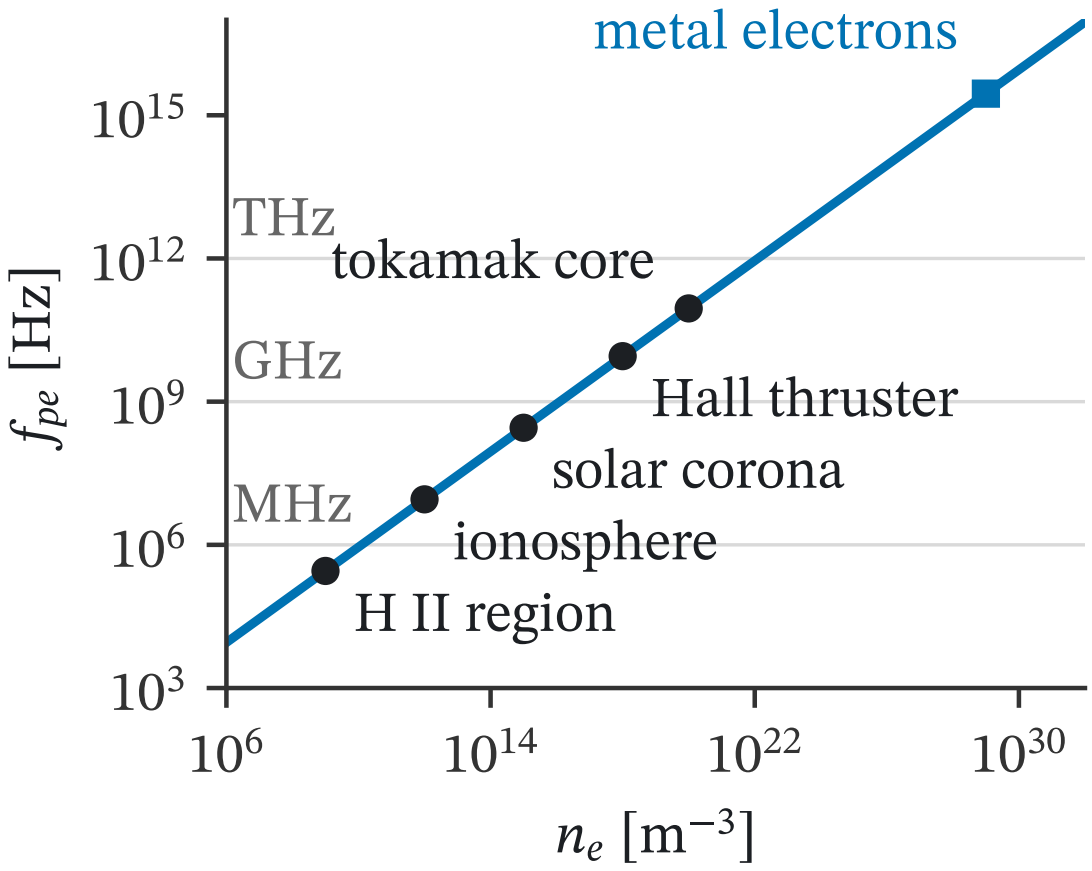
$$\omega_{pe} = \sqrt{\frac{n_0e^2}{\varepsilon_0 m_e}}$$

electron plasma frequency


$kv_0/\omega_{pe} = 0.5$
$\omega_{pe}t = 7.8$
measured $\omega/\omega_{pe} = 1.0000$
kv/ω_{pe} [1]
0.6
0
−0.6
sheets
$keE/(m_e\omega_{pe}^2)$ [1]
0.6
0
−0.6
0
0.5
1
x/λ [1]


$kv_0/\omega_{pe} = 1.5$
$\omega_{pe}t = 2.0$
kv/ω_{pe} [1] crossing $\omega_{pe}t = 0.73$ | $\arcsin(\omega_{pe}/kv_0) = 0.73$
3.2
0
−3.2
sheets
$keE/(m_e\omega_{pe}^2)$ [1]
1.6
0
−1.6
0
0.5
1
x/λ [1]

3.1 Plasma frequency



metal

$n_e = 10^{29}$ m^{-3}

$\omega_{pe} = 1.8 \times 10^{16}$ s^{-1}

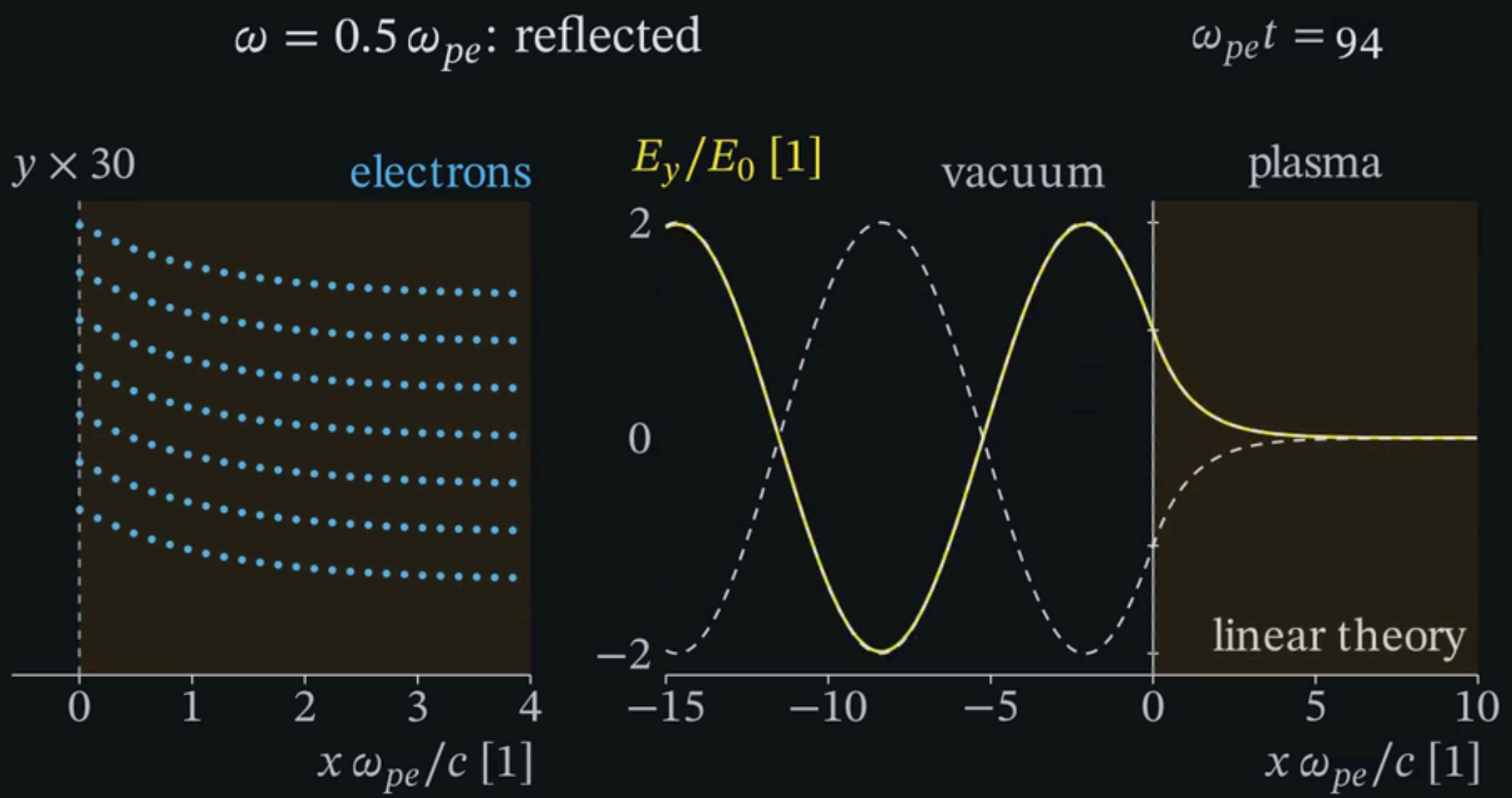
$\omega_{pe}^{-1} = 5.6 \times 10^{-17}$ s

fusion

$n_e = 10^{20}$ m^{-3}

$\omega_{pe} = 5.6 \times 10^{11}$ s^{-1}

$\omega_{pe}^{-1} = 1.8 \times 10^{-12}$ s


$\omega = 0.5\,\omega_{pe}$: reflected
$\omega_{pe}t = 94$
$y \times 30$
electrons
E_y/E_0 [1]
vacuum
plasma
2
0
−2
linear theory
0
1
2
3
4
$x\,\omega_{pe}/c$ [1]
−15
−10
−5
0
5
10
$x\,\omega_{pe}/c$ [1]

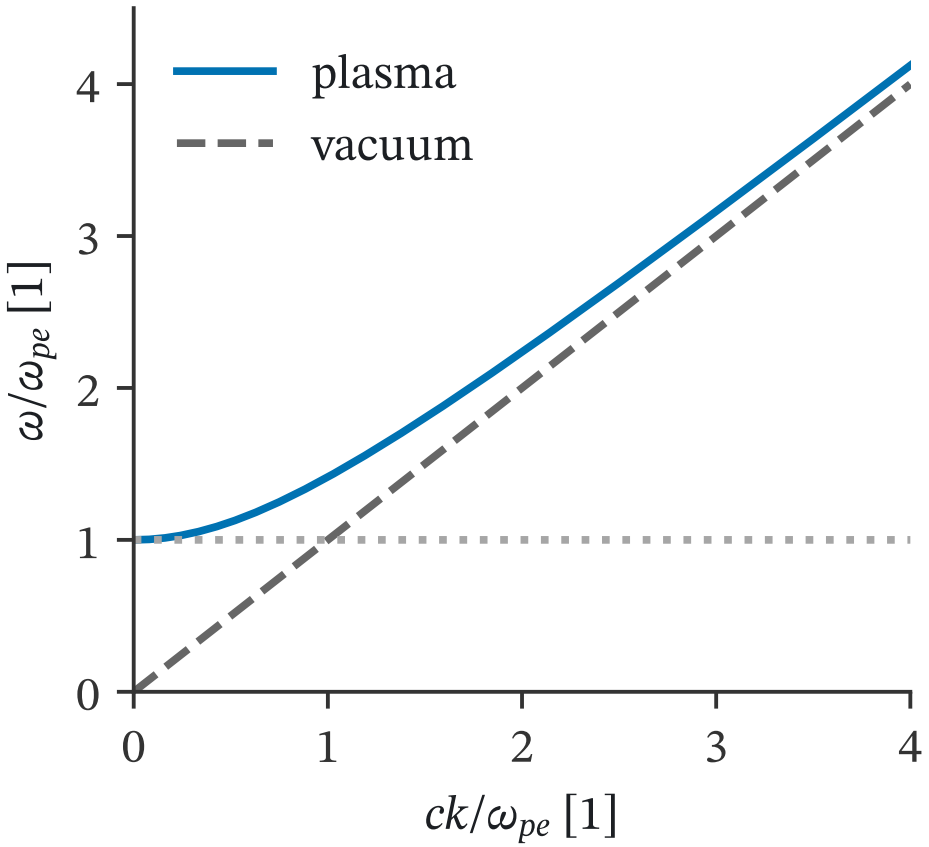
3.1 Electromagnetic waves

$$\omega^2 = \omega_{pe}^2 + c^2 k^2$$

dispersion relation

$\omega < \omega_{pe}$: reflected

$\omega > \omega_{pe}$: transmitted



3.1 Frequency and length scales

$$\omega_{ps} = \sqrt{\frac{n_s q_s^2}{\varepsilon_0 m_s}}$$

any species s

$$\omega_{pi} = \sqrt{\frac{Z m_e}{m_i}}\,\omega_{pe}$$

ions, $n_i = n_e/Z$: much slower

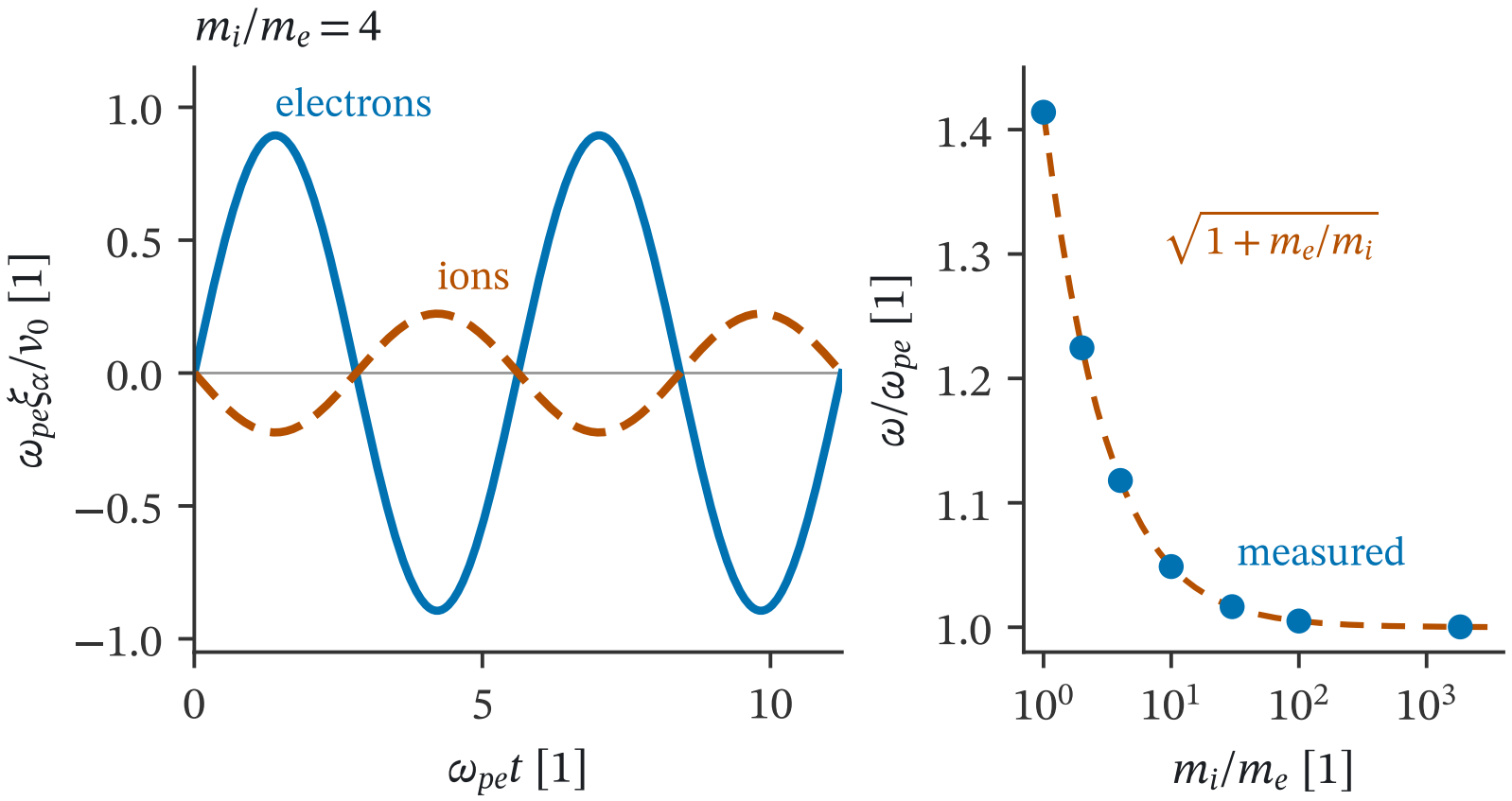
$$d_e = \frac{c}{\omega_{pe}}$$

inertial length: reach of a reflected wave

$$\omega_p^2 = \omega_{pe}^2 + \omega_{pi}^2 = \frac{n_e e^2}{\varepsilon_0 m_{\text{eff}}}$$

ions move too: $\frac{1}{m_{\text{eff}}} = \frac{1}{m_e} + \frac{Z}{m_i}$

3.1 Ions move too



$$\xi_i = -(m_e/m_i)\,\xi_e, \quad \omega_p^2 = \omega_{pe}^2 + \omega_{pi}^2, \quad \omega_{pi}^2/\omega_{pe}^2 = m_e/m_i$$

ξ_α displacement of species α, v_0 initial electron speed amplitude, $Z = 1$